



Convexity of ratios of the modified Bessel functions of the first kind with applications

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Abstract

Let $I_\nu(x)$ be the modified Bessel function of the first kind of order ν . Motivated by a conjecture on the convexity of the ratio $W_\nu(x) = xI_\nu(x)/I_{\nu+1}(x)$ for $\nu > -2$, using the monotonicity rules for a ratio of two power series and an elementary technique, we present fully the convexity of the functions $W_\nu(x)$, $W_\nu(x) - x^2/(2\nu + 4)$ and $W_\nu(x^{1/\theta})$ for $\theta \geq 2$ on $(0, \infty)$ in different value ranges of ν , which give an answer to the conjecture and extend known results. As consequences, some monotonicity results and new functional inequalities for $W_\nu(x)$ are established. As applications, an open problem and a conjectures are settled. Finally, a conjecture on the complete monotonicity of $W_\nu(x^{1/\theta})$ for $\theta \geq 2$ is proposed.

Keywords Modified Bessel function · Convexity · Monotonicity · Inequality

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Dedicated to people facing and battling COVID-19.

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1 Introduction

It is well-known that the second-order modified Bessel differential equation

$$x^2 y''(x) + xy'(x) - (x^2 + \nu^2) y(x) = 0, \quad (1.1)$$

has linearly independent solutions, that are, modified Bessel functions of the first and second, as usual, denoted by I_ν and K_ν , respectively, see [1, p. 77]. The modified Bessel function of the first kind $I_\nu(x)$ can be represented explicitly by the infinite series as

$$I_\nu(x) = \sum_{n=0}^{\infty} \frac{(x/2)^{2n+\nu}}{n! \Gamma(\nu + n + 1)}, \quad x \in \mathbb{R}, \nu \in \mathbb{R} \setminus \{-1, -2, \dots\}, \quad (1.2)$$

while the modified Bessel function of the second kind $K_\nu(x)$ is given by

$$K_\nu(x) = \frac{\pi}{2} \frac{I_{-\nu}(x) - I_\nu(x)}{\sin(\nu\pi)},$$

where the right-hand side of the formula above is replaced by its limiting value if ν is an integer. It is known that $I_\nu(x)$ satisfies the recurrence relations

$$xI'_\nu(x) + \nu I_\nu(x) = xI_{\nu-1}(x), \quad (1.3)$$

$$xI'_\nu(x) - \nu I_\nu(x) = xI_{\nu+1}(x), \quad (1.4)$$

which imply that

$$I_{\nu-1}(x) - I_{\nu+1}(x) = \frac{2\nu}{x} I_\nu(x) \quad (1.5)$$

and

$$y_\nu(x) := \frac{xI'_\nu(x)}{I_\nu(x)} = W_{\nu-1}(x) - \nu = \frac{x^2}{W_\nu(x)} + \nu, \quad (1.6)$$

where

$$W_\nu(x) = \frac{xI_\nu(x)}{I_{\nu+1}(x)} \text{ for } \nu > -2 \text{ and } x > 0. \quad (1.7)$$

The ratio $W_\nu(x) = xI_\nu(x)/I_{\nu+1}(x)$ plays an important role in finite elasticity [2, 3] and epidemiological models [4, 5], while another ratio $R_\nu(x) = I_{\nu+1}(x)/I_\nu(x)$ appeared in probability and statistics [6–8], and was applied in chemical kinetics [9, 10], optics [11] and signal processing [12]. For this reason, the properties of the ratio $W_\nu(x)$ have attracted many scholars' interest.

In 1983 Watson [13] showed that $W_\nu(x)$ is increasing on $(0, \infty)$ for all $\nu \geq 0$, which was used in the study of a Bayesian estimation problem by Robert [7]. This result was proven again in 1984 by Simpson and Spector [2] using another way. Baricz and Neuman [14, Theorem 2.2 (d)] in 2007 proved that $W_\nu(x)$ is increasing on $(0, \infty)$ for all $\nu > -2$. Another proof of this monotonicity property for $\nu > -1$ can be seen

in [15, Theorem 2.2 (i)]. In this paper, Baricz [15, Theorem 2.2 (i)] proved that the function $x \mapsto x^{\mu-\nu} I_\nu(x) / I_\mu(x)$ is strictly increasing on $(0, \infty)$ for all $\mu > \nu > -1$.

In [13] Watson also showed that the ratio $R_\nu(x) = I_{\nu+1}(x) / I_\nu(x)$ is increasing and concave on $(0, \infty)$ for all $\nu \geq -1/2$. Using this result, Baricz [16, Eq. (2.12)] proved that the ratio $y_\nu(x) (= W_{\nu-1}(x) - \nu)$ satisfies $y''_\nu(x) < 1/(\nu+1)$ for $\nu \geq -1/2$, which implies that the function $x \mapsto W_{\nu-1}(x) - x^2/(2\nu+2)$ is strictly concave on $(0, \infty)$ for $\nu \geq -1/2$. In 1984 Simpson and Spector [2] proved that the ratio $W_\nu(x)$ is convex on $(0, \infty)$ for all $\nu \geq 0$. Based on numerical experiments, Baricz [15, p. 591] made the following conjecture.

Conjecture 1 *The function $W_\nu(x)$ is strictly convex on $(0, \infty)$ for all $\nu > -1$.*

Remark 1 In a recent paper [17], Yang and Tian pointed out that Baricz's conjecture may be modified as follows: $W_\nu(x)$ is strictly convex on $(0, \infty)$ for all $\nu \geq -1/2$ and concave on $(0, \infty)$ if $-3/2 \leq \nu < -1/2$.

It is worth mentioning that, as early as in 1979, Ismail and Kelker [18] showed a more deep result which states that the function $x \mapsto x^{(\nu-\mu)/2} I_\mu(\sqrt{x}) / I_\nu(\sqrt{x})$ is strictly completely monotonic on $(0, \infty)$ for all $\mu > \nu > -1$. This, by choosing $\mu = \nu + 1$, implies that the function $x \mapsto 1/W_\nu(\sqrt{x})$ is strictly decreasing and convex on $(0, \infty)$ for $\nu > -1$.

Motivated by Baricz's conjecture and the consequence of Ismail and Kelker's result, the aim of this paper is to further investigate the convexity of the ratios $W_\nu(x)$ and $W_\nu(x^{1/\theta})$ on $(0, \infty)$ for $\nu > -2$ instead of $\nu > -1$ with $\theta \geq 2$.

The rest of this paper is organized as follows. In the next section, we recall some tools which will be used in this paper and establish an asymptotic expansion of $W_\nu(x)$. Our main results are presented in Sect. 3. In the first subsection, we present the monotonicity of the function $x W'_\nu(x) - \theta W_\nu(x)$ for $\theta = 1$ and $\theta \geq 2$; in the second subsection, we find that the ratio $W_\nu(x)$ is strictly convex on $(0, \infty)$ if and only if $\nu \geq -1/2$, which gives answer to Baricz's conjecture; while the functions $x \mapsto W_\nu(x^{1/\theta})$ for $\theta \geq 2$ and $x \mapsto W_\nu(x) - x^2/(2\nu+4)$ are both strictly concave on $(0, \infty)$ for $\nu > -2$, the former concavity improves a consequence of Ismail and Kelker; in the third subsection, we give necessary and sufficient conditions for which the functions $x \mapsto [W_\nu(x) - p]/x$ where $p = \nu + 1/2$ and $2(\nu+1)$ are respectively convex and concave on $(0, \infty)$. Some consequences of main results are given in Sect. 4, which include the monotonicity of three functions involving $W_\nu(x)$ and several new functional inequalities for $W_\nu(x)$. In Sect. 5, as applications, we give answers to Gaunt's open problem and Baricz–Neuman's conjecture. Finally, we make a conjecture that the function $x \mapsto W_\nu(x^{1/\theta})$ for $\theta \geq 2$ and $\nu > -1$ is a Bernstein function.

2 Preliminary results

To study the convexity of $W_\nu(x)$ on $(0, \infty)$ for all $\nu > -2$, we introduce the function

$$F_{\theta,\nu}(x) = x W'_\nu(x) - \theta W_\nu(x) \quad (2.1)$$

on $(0, \infty)$ for $\nu > -2$. An obvious fact is that

$$F'_{\theta, \nu}(x) = x W''_{\nu}(x) - (\theta - 1) W'_{\nu}(x) = x^{\theta} \left[x^{1-\theta} W'_{\nu}(x) \right]'. \quad (2.2)$$

In particular, $F'_{1, \nu}(x) = x W''_{\nu}(x)$. Thus, to show the convexity of $W_{\nu}(x)$, it suffices to investigate the monotonicity of the function $F_{1, \nu}(x)$. It was shown in [2] that the ratio $W_{\nu}(x)$ satisfies the Riccati equation

$$x W'_{\nu}(x) = x^2 + 2(\nu + 1) W_{\nu}(x) - W_{\nu}^2(x). \quad (2.3)$$

Thus, $F_{\theta, \nu}(x)$ can be expressed as

$$F_{\theta, \nu}(x) = x^2 + 2(\nu + 1 - \theta/2) W_{\nu}(x) - W_{\nu}^2(x) = -S_{\nu+1-\theta/2, \nu}(x), \quad (2.4)$$

where

$$S_{p, \nu}(x) = W_{\nu}^2(x) - 2p W_{\nu}(x) - x^2. \quad (2.5)$$

The monotonicity of $S_{p, \nu}(x)$ on $(0, \infty)$ for $\nu \geq -3/2$ was shown in [19]. For the sake of convenience, we quote some relevant computed results here. Let

$$f_1(x) = x^2 I_{\nu}(x)^2 - 2px I_{\nu}(x) I_{\nu+1}(x) - x^2 I_{\nu+1}(x)^2 \quad \text{and} \quad f_2(x) = I_{\nu+1}(x)^2. \quad (2.6)$$

It was shown in [19] that

$$S_{p, \nu}(x) = \frac{f_1(x)}{f_2(x)} = \frac{\sum_{n=0}^{\infty} a_n (x^2/4)^n}{\sum_{n=0}^{\infty} b_n (x^2/4)^n}, \quad (2.7)$$

where

$$a_n = 4 \frac{(2\nu - 2p + 1)n + (2\nu + 1)(\nu + 1 - p)}{(2n + 2\nu + 1)(n + \nu + 1)} \frac{\Gamma(\nu + 1)^2 \Gamma(2n + 2\nu + 2)}{n! \Gamma(n + \nu + 1)^2 \Gamma(n + 2\nu + 2)}, \quad (2.8)$$

$$b_n = \frac{2}{(n + \nu + 1)(n + 2\nu + 2)} \frac{\Gamma(\nu + 1)^2 \Gamma(2n + 2\nu + 2)}{n! \Gamma(n + \nu + 1)^2 \Gamma(n + 2\nu + 2)}, \quad (2.9)$$

and

$$\frac{a_n}{b_n} = 2 \frac{n + 2\nu + 2}{2n + 2\nu + 1} [(2\nu - 2p + 1)n + (2\nu + 1)(\nu + 1 - p)]. \quad (2.10)$$

It follows easily that

$$d_n := \frac{a_{n+1}}{b_{n+1}} - \frac{a_n}{b_n} = -2[p - h_n(\nu)], \quad (2.11)$$

where

$$h_n(\nu) = (2\nu + 1) \frac{2n^2 + 4(\nu + 1)n + \nu(2\nu + 3)}{(2n + 2\nu + 1)(2n + 2\nu + 3)}.$$

Moreover, it has been shown in [19] that

$$S_{p,\nu}(0) = \lim_{x \rightarrow 0} \frac{f_1(x)}{f_2(x)} = \frac{a_0}{b_0} = 4(\nu + 1)(\nu + 1 - p), \quad (2.12)$$

$$S_{\nu+1/2,\nu}(\infty) = \lim_{x \rightarrow \infty} \frac{f_1(x)}{f_2(x)} = \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \nu + \frac{1}{2}. \quad (2.13)$$

2.1 Three tools

To prove the monotonicity of $F_{\theta,\nu}(x)$, we need three tools. The first tool is the so-called H -function $H_{f,g}$ and two identities, which appeared in [20] and was called Yang's H -function and Yang's identities in [21] by Tian et. al. For $-\infty \leq a < b \leq \infty$, let f and g be differentiable on (a, b) and $g' \neq 0$ on (a, b) . Then the function $H_{f,g}$ is defined by

$$H_{f,g} := \frac{f'}{g'}g - f. \quad (2.14)$$

Using Yang's H -function, we have the following simple but useful identities:

$$\left(\frac{f}{g}\right)' = \frac{g'}{g^2} \left(\frac{f'}{g'}g - f\right) = \frac{g'}{g^2} H_{f,g}, \quad (2.15)$$

$$H'_{f,g} := \left(\frac{f'}{g'}\right)' g, \quad (2.16)$$

where f and g are twice differentiable on the interval (a, b) with $gg' \neq 0$ on (a, b) .

The second tool is the monotonicity rules for the ratio of two power series proven first by Biernacki and Krzyz [22], which play an important role in dealing with the monotonicity of the ratio of power series. A special monotonicity rule of two power series can be seen [23].

Lemma 1 ([22]) *Let a_n and b_n ($n = 0, 1, 2, \dots$) be real numbers and let the power series $A(t) = \sum_{n=1}^{\infty} a_n t^n$ and $B(t) = \sum_{n=1}^{\infty} b_n t^n$ be convergent for $|t| < r$. If $b_n > 0$ for $n = 0, 1, 2, \dots$, and a_n/b_n is strictly increasing (or decreasing) for $n = 0, 1, 2, \dots$, then the function $t \mapsto A(t)/B(t)$ is strictly increasing (or decreasing) on $(0, r)$.*

The third tool is the piecewise monotonicity rule presented by Yang, Chu and Wang in [24, Theorem 1], which is a nice complement to Biernacki and Krzyz's monotonicity rule. This new rule has been applied in the study for special functions, see for example, means and trigonometric [25–27], Bessel functions [28, 29], hypergeometric functions [30–35]. As a corollary of [24, Theorem 1], the following piecewise monotonicity rule on $(0, \infty)$, Lemma 2, first appeared in [36, Lemma 6.4] without proof and was also strictly proven in [37]. The present version of Lemma 2 was from [38] which modified something that is not rigorous in the assumptions for the sequences $\{a_k/b_k\}_{0 \leq k \leq m}$

and $\{a_k/b_k\}_{k \geq m}$ in the original version [24, Corollary 2.6] and modified version [19, Lemma 2.3].

Lemma 2 ([24, Corollary 2.6], [38]) *Let $A(t) = \sum_{k=0}^{\infty} a_k t^k$ and $B(t) = \sum_{k=0}^{\infty} b_k t^k$ be two real power series converging on $\mathbb{R}$ with $b_k > 0$ for all k . If for certain $m \in \mathbb{N}$, the non-constant sequences $\{a_k/b_k\}_{0 \leq k \leq m}$ and $\{a_k/b_k\}_{k \geq m}$ are respectively increasing (decreasing) and decreasing (increasing), then there is a unique $t_0 \in (0, \infty)$ such that the function A/B is increasing (decreasing) on $(0, t_0)$ and decreasing (increasing) on (t_0, ∞) .*

2.2 Asymptotic formulas involving $W_\nu(x)$ and $S_{p,\nu}(x)$

To investigate the convexity of $W_\nu(x)$ on $(0, \infty)$ for all $\nu > -2$, we also need the asymptotic expansions of $W_\nu(x)$ and $S_{p,\nu}(x)$. Using the asymptotic expansion (see [39, Eq. (10.40.1)])

$$I_\nu(x) \sim \frac{e^x}{\sqrt{2\pi x}} \sum_{n=0}^{\infty} \frac{\lambda_n(\nu)}{x^n} \text{ as } x \rightarrow \infty, \quad (2.17)$$

where $\lambda_0(\nu) = 1$ and for $k \geq 1$

$$\lambda_k(\nu) = \frac{(-1)^k}{k! 8^k} \prod_{j=1}^k \left(4\nu^2 - (2j-1)^2 \right), \quad (2.18)$$

we easily obtain

$$\frac{I_\nu(x)}{I_{\nu+1}(x)} \sim \sum_{n=0}^{\infty} \frac{c_n}{x^n}, \text{ as } x \rightarrow \infty,$$

with $c_0 = 1$ and for $n \geq 1$,

$$c_n = \lambda_n(\nu) - \sum_{k=1}^n c_{n-k} \lambda_k(\nu+1). \quad (2.19)$$

Then we obtain the following

Lemma 3 *The asymptotic formula*

$$W_\nu(x) = \frac{x I_\nu(x)}{I_{\nu+1}(x)} \sim \sum_{n=0}^{\infty} \frac{c_n}{x^{n-1}}, \text{ as } x \rightarrow \infty, \quad (2.20)$$

holds. In Particular, as $x \rightarrow \infty$,

$$W_\nu(x) \sim x + \nu + \frac{1}{2} + \frac{(2\nu+1)(2\nu+3)}{8x} + \frac{(2\nu+1)(2\nu+3)}{8x^2}, \quad (2.21)$$

$$W'_\nu(x) \sim 1 - \frac{(2\nu+1)(2\nu+3)}{8x^2} - \frac{(2\nu+1)(2\nu+3)}{4x^3}, \quad (2.22)$$

$$W''_\nu(x) \sim \frac{(2\nu+1)(2\nu+3)}{4x^3} + \frac{3(2\nu+1)(2\nu+3)}{4x^4}. \quad (2.23)$$

Remark 2 It follows that

$$\lim_{x \rightarrow \infty} (W_\nu(x) - x) = \nu + \frac{1}{2}, \quad \lim_{x \rightarrow \infty} W'_\nu(x) = 1, \quad \lim_{x \rightarrow \infty} W''_\nu(x) = 0. \quad (2.24)$$

By (1.2) we see that

$$I_\nu(x) \sim \frac{(x/2)^\nu}{\Gamma(\nu+1)} \left(1 + \frac{1}{4} \frac{x^2}{\nu+1} \right), \quad \text{as } x \rightarrow 0,$$

which leads to the following.

Lemma 4 We have

$$W_\nu(x) = \frac{x I_\nu(x)}{I_{\nu+1}(x)} \sim 2(\nu+2) \frac{x^2 + 4\nu + 4}{x^2 + 4\nu + 8}, \quad \text{as } x \rightarrow 0, \quad (2.25)$$

$$W'_\nu(x) \sim 16(\nu+2) \frac{x}{(x^2 + 4\nu + 8)^2} \rightarrow 0 \quad \text{as } x \rightarrow 0, \quad (2.26)$$

$$W''_\nu(x) \sim 16(\nu+2) \frac{4\nu + 8 - 3x^2}{(4\nu + 8 + x^2)^3} \rightarrow \frac{1}{\nu+2} \quad \text{as } x \rightarrow 0. \quad (2.27)$$

Remark 3 Using L'Hospital rule we have

$$\lim_{x \rightarrow 0} \frac{W_\nu(x) - 2(\nu+1)}{x^2/2} = \lim_{x \rightarrow 0} \frac{W'_\nu(x)}{x} = \lim_{x \rightarrow 0} W''_\nu(x) = \frac{1}{\nu+2}. \quad (2.28)$$

Lemma 5 Let $p = \nu + 1/2$. As $x \rightarrow \infty$, we have

$$\left[\frac{f_1(x)}{f_2(x)} \right]' \sim -\frac{(2\nu+3)(2\nu+1)}{4x^2}. \quad (2.29)$$

And therefore, if $-2 < \nu < -3/2$ then

$$H_{f_1, f_2}(x) = \frac{f'_1(x)}{f'_2(x)} f_2(x) - f_1(x) \rightarrow -\infty \quad \text{as } x \rightarrow \infty. \quad (2.30)$$

Proof We give first the asymptotic formula of the function

$$S_{p,\nu}(x) = W_{\nu}^2(x) - 2pW_{\nu}(x) - x^2 = \frac{f_1(x)}{f_2(x)}.$$

Using the asymptotic formula (2.21) yields, as $x \rightarrow \infty$,

$$\begin{aligned} \frac{f_1(x)}{f_2(x)} &\sim \left(x + \nu + \frac{1}{2} + \frac{(2\nu+1)(2\nu+3)}{8x} + \frac{(2\nu+1)(2\nu+3)}{8x^2} \right)^2 \\ &\quad - 2p \left(x + \nu + \frac{1}{2} + \frac{(2\nu+1)(2\nu+3)}{8x} + \frac{(2\nu+1)(2\nu+3)}{8x^2} \right) - x^2 \\ &\sim (2\nu+1-2p)x + (2\nu+1)(\nu+1-p) - \frac{(2\nu+3)(2\nu+1)(2p-2\nu-3)}{8x}. \end{aligned}$$

Particularly, when $p = \nu + 1/2$,

$$\frac{f_1(x)}{f_2(x)} \sim \nu + \frac{1}{2} + \frac{(2\nu+3)(2\nu+1)}{4x} \text{ as } x \rightarrow \infty,$$

and it is deduced that

$$\left[\frac{f_1(x)}{f_2(x)} \right]' \sim -\frac{(2\nu+3)(2\nu+1)}{4x^2} \text{ as } x \rightarrow \infty.$$

Since as $x \rightarrow \infty$,

$$\begin{aligned} f_2(x) &= I_{\nu+1}(x)^2 \sim \frac{e^{2x}}{2\pi x} \text{ and } f_2'(x) \sim \left[\frac{e^{2x}}{2\pi x} \right]' = \frac{(2x-1)e^{2x}}{2\pi x^2}, \\ \frac{f_2(x)^2}{f_2'(x)} &\sim \left(\frac{e^{2x}}{2\pi x} \right)^2 \bigg/ \frac{(2x-1)e^{2x}}{2\pi x^2} \sim \frac{e^{2x}}{4\pi x}. \end{aligned}$$

From the formula (2.15) we arrive at

$$H_{f_1, f_2}(x) = \frac{f_2(x)^2}{f_2'(x)} \left[\frac{f_1(x)}{f_2(x)} \right]' \sim -\frac{e^{2x}}{4\pi x} \frac{(2\nu+3)(2\nu+1)}{4x^2} \rightarrow -\infty \text{ as } x \rightarrow \infty,$$

if $-2 < \nu < -3/2$, which proves (2.30) and the proof is done. $\square$

3 Main results

We state and prove our main results in this section beginning with giving the monotonicity properties of the function $x \mapsto F_{\theta, \nu}(x)$ on $(0, \infty)$ for $\theta = 1$ and $\theta \geq 2$.

3.1 The monotonicity of $F_{\theta, \nu}(x) = xW'_\nu(x) - \theta W_\nu(x)$

We are now in a position to prove the monotonicity of $F_{\theta, \nu}(x) = xW'_\nu(x) - \theta W_\nu(x)$.

Theorem 1 *Let $\nu > -2$. The following statements are valid.*

- (i) *If $\nu \geq -1/2$, then the function $F_{1, \nu}(x) = xW'_\nu(x) - W_\nu(x)$ is increasing from $(0, \infty)$ onto $(-2\nu - 2, -\nu - 1/2)$, and therefore, the double inequality*

$$-2(\nu + 1) < xW'_\nu(x) - W_\nu(x) < -\left(\nu + \frac{1}{2}\right) \quad (3.1)$$

holds for $x > 0$.

- (ii) *If $-3/2 \leq \nu < -1/2$, then there exists an $x_0 > 0$ such that $F_{1, \nu}(x)$ is increasing on $(0, x_0)$ and decreasing on (x_0, ∞) . Moreover, the inequalities*

$$-2(\nu + 1) < xW'_\nu(x) - W_\nu(x) \leq F_{1, \nu}(x_0) < -(\nu + 1) \quad (3.2)$$

hold for $x > 0$.

- (iii) *If $-2 < \nu < -3/2$, then there exist $x_2 > x_1 > 0$ such that $F_{1, \nu}(x)$ is first increasing on $(0, x_1)$ then decreasing on (x_1, x_2) and finally increasing on (x_2, ∞) . Consequently, the inequalities*

$$\begin{aligned} -\frac{2(2\nu + 1)(\nu + 2)}{2\nu + 5} &< \min\{-2(\nu + 1), F_{1, \nu}(x_2)\} \leq xW'_\nu(x) - W_\nu(x) \\ &\leq \max\{-(\nu + 1/2), F_{1, \nu}(x_1)\} < -(\nu + 1) \end{aligned} \quad (3.3)$$

hold for $x > 0$.

Proof Using the relation (2.4) we have $F_{1, \nu}(x) = -S_{\nu+1/2, \nu}(x)$.

(i) When $\nu \geq -1/2$, by [19, Theorem 3.1 (ii)], we see that the function $S_{\nu+1/2, \nu}(x)$ is decreasing from $(0, \infty)$ onto $(\nu + 1/2, 2(\nu + 1))$, which implies the first assertion.

(ii) When $-3/2 \leq \nu < -1/2$, it was shown in [19, Theorem 3.4 (iii)] that, for $-3/2 < \nu < -1/2$, there exists an $x_0 > 0$ such that $S_{\nu+1/2, \nu}(x)$ is decreasing on $(0, x_0)$ and increasing on (x_0, ∞) with the estimates

$$S_{\nu+1/2, \nu}(x_0) \leq S_{\nu+1/2, \nu}(x) < 2\nu + 2$$

for all $x > 0$, where x_0 is a unique solution of the equation $S'_{\nu+1/2, \nu}(x) = 0$ on $(0, \infty)$, and then the monotonicity of the function $F_{1, \nu}(x) = -S_{\nu+1/2, \nu}(x)$ follows. It is easy to check that this conclusion is also true for $\nu = -3/2$ (the same below). To prove the third inequality of (3.2), it suffices to prove $S_{\nu+1/2, \nu}(x_0) > 2\nu + 1$. Substituting $p = \nu + 1/2$ into the formula (2.10) yields

$$\frac{a_n}{b_n} = (2\nu + 1) \frac{n + 2\nu + 2}{2n + 2\nu + 1}, \quad (3.4)$$

and then

$$\frac{a_n}{b_n} - (2\nu + 1) = (2\nu + 1) \frac{n + 2\nu + 2}{2n + 2\nu + 1} = -\frac{(2\nu + 1)(n - 1)}{2n + 2\nu + 1} \geq 0 \quad (3.5)$$

for all $n \geq 0$ and $-2 < \nu < -1/2$. Since $b_n > 0$ for all $n \geq 0$, it follows from the formula (2.7) that

$$\begin{aligned} S_{\nu+1/2, \nu}(x) - (2\nu + 1) &= \frac{\sum_{n=0}^{\infty} a_n (x^2/4)^n}{\sum_{n=0}^{\infty} b_n (x^2/4)^n} - (2\nu + 1) \\ &= \frac{\sum_{n=0}^{\infty} [a_n/b_n - (2\nu + 1)] b_n (x^2/4)^n}{\sum_{n=0}^{\infty} b_n (x^2/4)^n} > 0 \end{aligned} \quad (3.6)$$

for all $x > 0$ and $-2 < \nu < -1/2$, which completes the proof of the second assertion of this theorem.

(iii) When $-2 < \nu < -3/2$, since $F_{1, \nu}(x) = -S_{\nu+1/2, \nu}(x)$, it suffices to prove that there exist $x_2 > x_1 > 0$ such that $S_{p, \nu}(x) = f_1(x)/f_2(x)$ for $p = \nu + 1/2$ is first decreasing on $(0, x_1)$ then increasing on (x_1, x_2) and finally decreasing on (x_2, ∞) . Substituting $p = \nu + 1/2$ into (2.11) gives

$$d_n = \frac{a_{n+1}}{b_{n+1}} - \frac{a_n}{b_n} = -2 \left[\nu + \frac{1}{2} - h_n(\nu) \right] = -\frac{(2\nu + 1)(2\nu + 3)}{(2n + 2\nu + 1)(2n + 2\nu + 3)}.$$

A simple computation yields $d_0 = -1 < 0$, $d_1 = -(2\nu + 1)/(2\nu + 5) > 0$ and $d_n < 0$ for $n \geq 2$, which implies that the sequence $\{a_n/b_n\}$ is decreasing for $n = 0, 1$, then increasing for $n = 1, 2$, and decreasing for $n \geq 2$. Clearly, Lemmas 1 and 2 do not directly apply in such case. For this, we let

$$f_i(x) = g_i\left(\frac{x^2}{4}\right), \quad i = 1, 2.$$

By (2.7), we have

$$\frac{g_1(t)}{g_2(t)} = \frac{\sum_{n=0}^{\infty} a_n t^n}{\sum_{n=0}^{\infty} b_n t^n}.$$

Differentiation yields

$$\frac{g'_1(t)}{g'_2(t)} = \frac{\sum_{n=1}^{\infty} n a_n t^{n-1}}{\sum_{n=1}^{\infty} n b_n t^{n-1}} = \frac{\sum_{n=0}^{\infty} (n+1) a_{n+1} t^n}{\sum_{n=0}^{\infty} (n+1) b_{n+1} t^n}.$$

Obviously, the sequence $\{(n+1)a_{n+1}/(n+1)b_{n+1}\} = \{a_{n+1}/b_{n+1}\}$ is increasing for $n = 0, 1$ and decreasing for $n \geq 1$. By Lemma 2, there is a unique $t_0 \in (0, \infty)$ such that the function $g_1(t)/g_2(t)$ is increasing on $(0, t_0)$ and decreasing on (t_0, ∞) .

Using the formula (2.16), we see that $H'_{g_1, g_2} = (g'_1/g'_2)' g_2 > 0$ for $t \in (0, t_0)$ and $H'_{g_1, g_2} < 0$ for $t \in (t_0, \infty)$ due to $g_2 > 0$. Since

$$H_{g_1, g_2}(0) = \frac{g'_1(0)}{g'_2(0)} g_2(0) - g_1(0) = \frac{a_1}{b_1} b_0 - a_0 = -\frac{1}{(\nu+1)^2} < 0,$$

and by Lemma 5,

$$\lim_{t \rightarrow \infty} H_{g_1, g_2}(t) = \lim_{t \rightarrow \infty} \left[\frac{g'_1(t)}{g'_2(t)} g_2(t) - g_1(t) \right] = \lim_{x \rightarrow \infty} \left[\frac{f'_1(x)}{f'_2(x)} f_2(x) - f_1(x) \right] = -\infty,$$

where $x = 2\sqrt{t}$, we claim that $H_{g_1, g_2}(t_0) > 0$. If not, that is, $H_{g_1, g_2}(t_0) \leq 0$, then $H_{g_1, g_2}(t) \leq 0$ for all $t > 0$. This, in view of $g'_2 > 0$, in combination with the relation $(g_1/g_2)' = (g'_2/g_2^2) H_{g_1, g_2}$ yields $(g_1/g_2)' \leq 0$, that is, g_1/g_2 is decreasing on $(0, \infty)$. It thus follows that

$$2(\nu+1) = \frac{a_0}{b_0} = \lim_{t \rightarrow 0} \frac{g_1(t)}{g_2(t)} \geq \lim_{t \rightarrow \infty} \frac{g_1(t)}{g_2(t)} = \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \nu + \frac{1}{2},$$

which implies that $\nu \geq -3/2$. This is in contradiction with the assumption that $-2 < \nu < -3/2$. Consequently, there are two positive numbers t_1 and t_2 with $t_1 < t_2$ for which $H_{g_1, g_2}(t) < 0$ for $t \in (0, t_1) \cup (t_2, \infty)$ and $H_{g_1, g_2}(t) > 0$ for $t \in (t_1, t_2)$. Then $(g_1(t)/g_2(t))' < 0$ for $t \in (0, t_1) \cup (t_2, \infty)$ and $(g_1(t)/g_2(t))' > 0$ for $t \in (t_1, t_2)$, and therefore, $(f_1(x)/f_2(x))' < 0$ for $x \in (0, x_1) \cup (x_2, \infty)$ and $(f_1(x)/f_2(x))' > 0$ for $x \in (x_1, x_2)$, where $x_i = 2\sqrt{t_i}$, $i = 1, 2$, which proves the required piecewise monotonicity. From the monotonicity of the function $S_{p, \nu}(x) = f_1(x)/f_2(x)$ for $p = \nu + 1/2$ on $(0, \infty)$ we have

$$\min \{S_{\nu+1/2, \nu}(x_1), S_{\nu+1/2, \nu}(\infty)\} \leq S_{\nu+1/2, \nu}(x) \leq \max \{S_{\nu+1/2, \nu}(0), S_{\nu+1/2, \nu}(x_2)\}$$

for all $x > 0$. Since the inequality (3.5) holds for all $n \geq 0$ and $\nu < -1/2$, It thus holds that

$$\min \{S_{\nu+1/2, \nu}(x_1), S_{\nu+1/2, \nu}(\infty)\} > \min \{2\nu+1, \nu+1/2\} = 2\nu+1.$$

Finally, using the formula (3.4) leads to

$$\frac{a_n}{b_n} - \frac{2(2\nu+1)(\nu+2)}{2\nu+5} = -\frac{(2\nu+1)(2\nu+3)(n-2)}{(2\nu+5)(2n+2\nu+1)} \leq 0$$

for all $n \geq 0$ and $-2 < \nu < -3/2$. It then follows that

$$\begin{aligned} S_{\nu+1/2,\nu}(x) - \frac{2(2\nu+1)(\nu+2)}{2\nu+5} &= \frac{\sum_{n=0}^{\infty} a_n (x^2/4)^n}{\sum_{n=0}^{\infty} b_n (x^2/4)^n} - (2\nu+1) \\ &= \frac{1}{\sum_{n=0}^{\infty} b_n (x^2/4)^n} \sum_{n=0}^{\infty} \left[\frac{a_n}{b_n} - \frac{2(2\nu+1)(\nu+2)}{2\nu+5} \right] b_n (x^2/4)^n < 0 \end{aligned}$$

for all $x > 0$ and $-2 < \nu < -3/2$, which indicates that

$$\begin{aligned} \max \{S_{\nu+1/2,\nu}(0), S_{\nu+1/2,\nu}(x_2)\} &< \max \left\{ 2\nu+2, \frac{2(2\nu+1)(\nu+2)}{2\nu+5} \right\} \\ &= \frac{2(2\nu+1)(\nu+2)}{2\nu+5}. \end{aligned}$$

Due to $F_{1,\nu}(x) = xW'_\nu(x) - W_\nu(x) = -S_{\nu+1/2,\nu}(x)$, the inequalities (3.3) follow, and the proof is completed. $\square$

Theorem 2 Let $\theta \geq 2$ and $\nu > -2$. The function $F_{\theta,\nu}(x) = xW'_\nu(x) - \theta W_\nu(x)$ is decreasing from $(0, \infty)$ onto $(-\infty, -2\theta(\nu+1))$, and hence, the inequality

$$xW'_\nu(x) - \theta W_\nu(x) < -2\theta(\nu+1) \quad (3.7)$$

holds for $x > 0$.

Proof From [19, Theorem 3.1 (iv) and Theorem 3.4 (v)] we see that if $p = \nu + 1 - \theta/2 \leq \nu$, that is, $\theta \geq 2$, then $F_{\theta,\nu}(x) = -S_{\nu+1-\theta/2,\nu}(x)$ is decreasing on $(0, \infty)$ for $\nu \geq -3/2$. We prove that this conclusion is also valid for all $\nu > -2$. In fact, since $b_0 = 1/(\nu+1)^2 > 0$ and $b_n > 0$ for all $n \geq 1$ and $\nu > -2$, if $p \leq \nu$ then

$$\frac{a_{n+1}}{b_{n+1}} - \frac{a_n}{b_n} \geq -2[\nu - h_n(\nu)] = \frac{4n(n+2\nu+2)}{(2n+2\nu+1)(2n+2\nu+3)} \geq 0$$

for all $n \geq 0$ and $\nu > -2$, that is, the sequence $\{a_n/b_n\}_{n \geq 0}$ is increasing. By Lemma 1 the function $S_{p,\nu}(x)$ for $p \leq \nu$ is increasing on $(0, \infty)$, which implies that $F_{\theta,\nu}(x) = -S_{\nu+1-\theta/2,\nu}(x)$ is decreasing on $(0, \infty)$. The desired inequality follows from $F'_{\theta,\nu}(x) < 0$ for $x > 0$. This completes the proof. $\square$

3.2 The convexity of $W_\nu(x)$

We are now in a position to state and prove the convexity of the ratio W_ν . First, we prove the following theorem, which describes fully the convex pattern of the ratio $W_\nu(x)$ and gives an answer to Baricz's conjecture stated in Introduction.

Theorem 3 Let $\nu > -2$. (i) The ratio $W_\nu(x)$ is strictly convex on $(0, \infty)$ if and only if $\nu \geq -1/2$; (ii) if $-3/2 \leq \nu < -1/2$, then $W_\nu(x)$ is convex then concave on $(0, \infty)$; (iii) while $-2 < \nu < -3/2$, $W_\nu(x)$ is first convex then concave and finally convex on $(0, \infty)$.

Proof (i) Using the asymptotic relation (2.23), the necessary condition for which $W_\nu(x)$ is strictly convex on $(0, \infty)$ is that: $\nu \geq -1/2$. Since $F_{1,\nu}(x) = xW'_\nu(x) - W_\nu(x)$ is strictly increasing on $(0, \infty)$ by Theorem 1 (i), we immediately get that $F'_{1,\nu}(x) = xW''_\nu(x) > 0$ for $x > 0$, which implies that the ratio $W_\nu(x)$ is strictly convex on $(0, \infty)$, and the sufficiency follows.

(ii) If $-3/2 \leq \nu < -1/2$, then by Theorem 1 (ii), there exists an $x_0 > 0$ such that $F'_{1,\nu}(x) = xW''_\nu(x) > 0$ for $x \in (0, x_0)$ and $F'_{1,\nu}(x) = xW''_\nu(x) < 0$ for $x \in (x_0, \infty)$, which implies the second required claim.

(iii) If $-2 < \nu < -3/2$, then using Theorem 1 (iii) we can prove the required assertion, and the proof is done. $\square$

Due to the relation (2.2), that is, $F'_{\theta,\nu}(x) = x^\theta [x^{1-\theta} W'_\nu(x)]'$, we can obtain an interesting concavity of the ratio $W_\nu(x^{1/\theta})$ for $\theta \geq 2$ and the function $x \mapsto W_\nu(x) - x^2/(2\nu + 4)$ on $(0, \infty)$ for $\nu > -2$.

Theorem 4 *Let $\theta \geq 2$ and $\nu \geq -2$. Then the following statements are valid.*

(i) *Then function $x \mapsto x^{1-\theta} W'_\nu(x)$ is strictly decreasing on $(0, \infty)$. Taking $\theta = 2$, the double inequality*

$$\max \{W''_\nu(x), 0\} < \frac{W'_\nu(x)}{x} < \frac{1}{\nu + 2}, \quad (3.8)$$

holds for $x > 0$.

(ii) *Both of the functions*

$$x \mapsto W_\nu(x^{1/\theta}) \quad \text{and} \quad x \mapsto W_\nu(x) - \frac{x^2}{2(\nu + 2)}$$

are strictly concave on $(0, \infty)$.

Proof (i) The decreasing property of the function $x \mapsto x^{1-\theta} W'_\nu(x)$ on $(0, \infty)$ follows from Theorem 2 and the relation (2.2). This property for $\theta = 2$ implies that

$$0 = \lim_{x \rightarrow \infty} \frac{W'_\nu(x)}{x} < \frac{W'_\nu(x)}{x} < \lim_{x \rightarrow 0} \frac{W'_\nu(x)}{x} = \frac{1}{\nu + 2}$$

for $x > 0$; while $F'_{2,\nu}(x) = xW''_\nu(x) - W_\nu(x) < 0$ for $x > 0$ implies that $W''_\nu(x) < W'_\nu(x)/x$ for $x > 0$. The double inequality (3.8) thus follows.

(ii) Differentiation leads to

$$\begin{aligned} \frac{d}{dx} W_\nu(x^{1/\theta}) &= \frac{1}{\theta} x^{1/\theta-1} W'_\nu(x^{1/\theta}) = \frac{1}{\theta} t^{1-\theta} W'_\nu(t), \text{ where } t = x^{1/\theta}, \\ \frac{d^2}{dx^2} W_\nu(x^{1/\theta}) &= \frac{t^{1-\theta}}{\theta^2} [t^{1-\theta} W'_\nu(t)]' < 0 \end{aligned}$$

for $x > 0$. Using the inequalities (3.8) yields

$$\left[W_v(x) - \frac{x^2}{2(v+2)} \right]'' = W_v''(x) - \frac{1}{v+2} < 0$$

for $x > 0$, thereby completing the proof. $\square$

Remark 4 From Theorem 4 (ii) we immediately deduce that the functions

$$x \mapsto \frac{1}{W_v(x^{1/\theta})} \text{ and } x \mapsto \ln W_v(x^{1/\theta})$$

are both strictly convex on $(0, \infty)$ for $\theta \geq 2$ and $v > -1$. In particular, the convexity of the ratio $1/W_v(\sqrt{x})$ is a direct consequence of Ismail and Kelker's deep result in [18].

Remark 5 It was shown in Baricz [16, Eq. (2.12)] that the inequalities $y_v''(x) < 1/(v+1)$ and $y_v'(x) < x/(v+1)$ holds for $x > 0$ and $v \geq -1/2$. Due to the relation 1.6, that is, $y_v(x) = W_{v-1}(x) - v$, Theorem 4 (ii) implies that the two inequalities hold for $x > 0$ and $v > -1$, which extends the value range of v from $v \geq -1/2$ to $v > -1$.

3.3 The convexity of $\xi_{p,v}(x) = [W_v(x) - p]/x$

Theorem 5 Let $p \in \mathbb{R}$, $v > -2$ and let the function $\xi_{p,v}$ be defined on $(0, \infty)$ by

$$\xi_{p,v}(x) = \frac{W_v(x) - p}{x}. \quad (3.9)$$

The function $\xi_{v+1/2,v}(x)$ is convex on $(0, \infty)$ if and only if $v \geq -1/2$, while $\xi_{2v+2,v}(x)$ is concave on $(0, \infty)$ if and only if $v \geq -3/2$.

Proof Differentiation yields

$$x^2 \xi'_{p,v}(x) = x W_v'(x) - W_v(x) + p, \quad (3.10)$$

$$x^3 \xi''_{p,v}(x) = x^2 W_v''(x) - 2[x W_v'(x) - W_v(x) + p]. \quad (3.11)$$

Using the asymptotic formulas (2.21)–(2.23) we have

$$x^3 \xi''_{p,v}(x) \sim 2v + 1 - 2p + \frac{3(2v+1)(2v+3)}{4x}, \text{ as } x \rightarrow \infty,$$

which implies that

$$\begin{aligned} \lim_{x \rightarrow \infty} [x^4 \xi''_{v+1/2,v}(x)] &= 3 \left(v + \frac{1}{2} \right) \left(v + \frac{3}{2} \right), \\ \lim_{x \rightarrow \infty} [x^3 \xi''_{2v+2,v}(x)] &= -(2v+3). \end{aligned}$$

(i) If $\xi_{\nu+1/2,\nu}(x)$ is convex on $(0, \infty)$, then $\lim_{x \rightarrow \infty} \left[x^4 \xi_{\nu+1/2,\nu}''(x) \right] \geq 0$, which implies that $\nu \geq -1/2$. Conversely, if $\nu \geq -1/2$, then by the right hand side inequality of (3.1),

$$x W_{\nu}'(x) - W_{\nu}(x) + \nu + \frac{1}{2} < 0$$

for $x > 0$, which, in combination with $W_{\nu}''(x) > 0$ for $x > 0$ by Theorem 3, yields

$$x^3 \xi_{\nu+1/2,\nu}''(x) = x^2 W_{\nu}''(x) - 2 \left[x W_{\nu}'(x) - W_{\nu}(x) + \nu + \frac{1}{2} \right] > 0$$

for $x > 0$, and therefore, $\xi_{\nu+1/2,\nu}''(x) > 0$ for $x > 0$.

(ii) If $\xi_{2\nu+2,\nu}(x)$ is concave on $(0, \infty)$, then $\lim_{x \rightarrow \infty} \left[x^3 \xi_{2\nu+2,\nu}''(x) \right] \leq 0$, which implies that $\nu \geq -3/2$. To prove the sufficiency, we use the relation (1.6), which implies

$$W_{\nu-1}(x) = \frac{x^2}{W_{\nu}(x)} + 2\nu,$$

to write $\xi_{2\nu+2,\nu}(x)$ as

$$\xi_{2\nu+2,\nu}(x) = \frac{W_{\nu}(x) - 2(\nu+1)}{x} = \frac{x}{W_{\nu+1}(x)}. \quad (3.12)$$

Differentiation again yields

$$\begin{aligned} \xi_{2\nu+2,\nu}'(x) &= \frac{W_{\nu+1}(x) - x W_{\nu+1}'(x)}{W_{\nu+1}(x)^2}, \\ \xi_{2\nu+2,\nu}''(x) &= \frac{-x W_{\nu+1}(x) W_{\nu+1}''(x) + 2 [x W_{\nu+1}'(x) - W_{\nu+1}(x)] W_{\nu+1}'(x)}{W_{\nu+1}(x)^3}. \end{aligned}$$

Since $W_{\nu+1}^{(i)}(x) > 0$ ($i = 0, 1, 2$) for $x > 0$ and $\nu+1 \geq -1/2$, and $x W_{\nu+1}'(x) - W_{\nu+1}(x) < -(\nu+3/2) \leq 0$ for $x > 0$ and $\nu+1 \geq -1/2$ due to the right hand side inequality of (3.1), we immediately get $\xi_{2\nu+2,\nu}''(x) < 0$ for $x > 0$ and $\nu \geq -3/2$. This completes the proof. $\square$

Remark 6 The relation (3.12) implies that $\xi_{2\nu+2,\nu}(x) = I_{\nu+2}(x)/I_{\nu+1}(x) = R_{\nu+1}(x)$. The second assertion of Theorem 5 means that the ratio $R_{\nu}(x)$ is concave on $(0, \infty)$ if and only if $\nu \geq -1/2$. This result was due to Watson [13].

4 Consequences

In this section, we give some direct consequences of the convexity of the ratios $W_{\nu}(x)$ and $W_{\nu}(x^{1/\theta})$ for $\theta \geq 2$, including some monotonicity and new functional inequalities for $W_{\nu}(x)$.

4.1 Three monotonicity results involving $W_\nu(x)$

First, we consider the difference $W_\nu(x) - x$, which appeared in the study of generalized Marcum Q-function, see [40, Lemma 1]. The monotonicity of $x \mapsto y_\nu(x) - x$ on $(0, \infty)$ for $\nu \geq 1/2$ was proven in [41], which, since $y_\nu(x) = W_{\nu-1}(x) - \nu$, implies that the difference $W_\nu(x) - x$ has the same monotonicity as the function $x \mapsto y_{\nu+1}(x) - x$ for $\nu \geq -1/2$. In the following corollary, we present a new proof of this monotonicity for $\nu \geq -1/2$ and discuss the piecewise monotonicity for $-2 < \nu < -1/2$ using the convexity of the ratio $W_\nu(x)$ on $(0, \infty)$ for $\nu > -2$.

Corollary 1 *Let the function δ_ν defined on $(0, \infty)$ for $\nu > -2$ by $\delta_\nu(x) = W_\nu(x) - x$. The following statements are valid:*

- (i) *If $\nu \geq -1/2$, then the function $\delta_\nu(x)$ is decreasing from $(0, \infty)$ onto $(\nu + 1/2, 2\nu + 2)$. Consequently, the double inequality*

$$x + \nu + \frac{1}{2} < W_\nu(x) < x + 2(\nu + 1) \quad (4.1)$$

holds for $x > 0$.

- (ii) *If $-3/2 \leq \nu < -1/2$, then the function $\delta_\nu(x)$ is decreasing then increasing on $(0, \infty)$, and therefore, the right hand side inequality of (4.1) holds for $x > 0$.*
- (iii) *If $-2 < \nu < -3/2$, then the function $\delta_\nu(x)$ is first decreasing then increasing and finally decreasing on $(0, \infty)$.*

Proof Using the relations (2.25) and (2.21) gives

$$\delta_\nu(0) = 2(\nu + 1) \text{ and } \delta_\nu(\infty) = \lim_{x \rightarrow \infty} \delta_\nu(x) = \nu + 1/2. \quad (4.2)$$

(i) In the case of $\nu \geq -1/2$, due to $W''_\nu(x) > 0$ for $x > 0$ by Theorem 3 (i), we derive that

$$0 = W'_\nu(0) < W'_\nu(x) < \lim_{x \rightarrow \infty} W'_\nu(x) = 1, \quad (4.3)$$

where $W'_\nu(0) = 0$ and $\lim_{x \rightarrow \infty} W'_\nu(x) = 1$ are as in (2.26) and (2.24). This implies that $\delta'_\nu(x) = W'_\nu(x) - 1 < 0$ for $x > 0$, that is, the function $\delta_\nu(x) = W_\nu(x) - x$ is strictly decreasing on $(0, \infty)$. By virtue of the decreasing property of $\delta_\nu(x)$ on $(0, \infty)$, the double inequality (4.1) follows.

(ii) In the case of $-3/2 \leq \nu < -1/2$, by Theorem 3 (i) there is an $x_2 > 0$ such that $\delta''_\nu(x) = W''_\nu(x) > 0$ for $x \in (0, x_2)$ and $\delta''_\nu(x) = W''_\nu(x) < 0$ for $x \in (x_2, \infty)$, that is, $\delta'_\nu(x)$ is increasing on $(0, x_2)$ and decreasing on (x_2, ∞) . This in conjunction with facts that

$$\delta'_\nu(x) = W'_\nu(x) - 1 = -1 < 0 \text{ and } \lim_{x \rightarrow \infty} \delta'_\nu(x) = \lim_{x \rightarrow \infty} W'_\nu(x) - 1 = 0$$

leads to that there is an $x_1 \in (x_2, \infty)$ for which $\delta'_\nu(x) < 0$ for $x \in (0, x_1)$ and $\delta'_\nu(x) > 0$ for $x \in (x_1, \infty)$. We thus obtain

$$\delta_\nu(x) < \max\{\delta_\nu(0), \delta_\nu(\infty)\} = \max\left\{2(\nu+1), \nu + \frac{1}{2}\right\} = 2(\nu+1),$$

which indicates that the right hand side inequality of (4.1) holds for $x > 0$.

(iii) In the case of $-2 < \nu < -3/2$, it is seen from Theorem 3 (ii) that there are $x_4 > x_3 > 0$ such that $\delta''_\nu(x) = W''_\nu(x) > 0$ for $x \in (0, x_3) \cup (x_4, \infty)$ and $\delta''_\nu(x) = W''_\nu(x) < 0$ for $x \in (x_3, x_4)$. Since $\delta'_\nu(0) = -1 < 0$ and $\delta'_\nu(\infty) = 0$, we have $\delta'_\nu(x_4) < \delta'_\nu(\infty) = 0$. Moreover, we have $\delta'_\nu(x_3) > 0$. If not, that is, $\delta'_\nu(x_3) \leq 0$, then $\delta'_\nu(x) \leq 0$ for all $x > 0$, which means that $\delta_\nu(x)$ is decreasing on $(0, \infty)$, and therefore,

$$2(\nu+1) = \delta_\nu(0) > \delta_\nu(\infty) = \nu + \frac{1}{2},$$

namely, $\nu > -3/2$. This is clearly in contradiction with the assumption that $-2 < \nu < -3/2$. Thus, there are two $x_1 \in (0, x_3)$ and $x_2 \in (x_3, x_4)$ such that $\delta'_\nu(x) < 0$ for $x \in (0, x_1) \cup (x_2, \infty)$ and $\delta'_\nu(x) > 0$ for $x \in (x_1, x_2)$.

We thus complete the proof. $\square$

Remark 7 Due to the relation (1.6), the first inequality of (4.1) is equivalent to

$$x - \frac{1}{2} < \frac{xI'_\nu(x)}{I_\nu(x)} = y_\nu(x)$$

for $x > 0$ and $\nu \geq 1/2$. This was proven in [41] by Gronwall. The second inequality of (4.1) was shown in [42] by Násell.

Remark 8 Using the monotonicity of the function $x \mapsto W_\nu(x) - x$, the proof of Baricz's Lemma 1 in [40] can be greatly simplified. Moreover, a version of the modified Bessel function of the second of Corollary 1 was shown in [43, Proposition 2] by Yang and Tian.

Second, using Theorems 5 and 1 we have the following monotonicity results of the function $\xi_{p,\nu}(x)$ defined by (3.9) on $(0, \infty)$ for $p = \nu + 1/2, 2(\nu + 1)$.

Corollary 2 Let $\nu > -2$. The following statements are valid:

- (i) The function $\xi_{\nu+1/2,\nu}(x)$ is strictly decreasing on $(0, \infty)$ if and only if $\nu \geq -1/2$; when $-3/2 < \nu < -1/2$, it is decreasing then increasing on $(0, \infty)$; while $-2 < \nu < -3/2$, it is increasing then decreasing on $(0, \infty)$; it is strictly increasing on $(0, \infty)$ if and only if $\nu = -3/2$.
- (ii) The function $\xi_{2\nu+2,\nu}(x)$ is strictly increasing if and only if $\nu \geq -3/2$, while $-2 < \nu < -3/2$, it is increasing then decreasing on $(0, \infty)$.

Using the concavity of the function $x \mapsto W_\nu(x) - x^2/(2\nu+4)$ on $(0, \infty)$ for $\nu > -2$ and the inequality (3.7) for $\theta = 2$, we have

Corollary 3 Let $\nu > -2$. The functions $x \mapsto W_\nu(x) - x^2/(2\nu + 4)$ and $x \mapsto [W_\nu(x) - 2(\nu + 1)]/x^2$ are both strictly decreasing from $(0, \infty)$ onto $(-\infty, 0)$. Consequently, the double inequality

$$2(\nu + 1) < W_\nu(x) < 2(\nu + 1) + \frac{x^2}{2(\nu + 2)} \quad (4.4)$$

holds for $x > 0$.

4.2 Several new inequalities for $W_\nu(x)$

There are various existing inequalities for the ratio $W_\nu(x)$ for $x > 0$ and different range of values of ν , which mainly include Amos type inequality that $W_\nu(x) < (>) p + \sqrt{x^2 + q^2}$ (see [44, 45]), Simpson-Spector type inequality that $W_\nu(x) < (>) S_{p,\nu}(x)$ (see [2, 19, 46]), Yang-Zheng type inequality that $W_\nu(x) < (>) p + r\sqrt{x^2 + q^2}$ ($r > 0$) (see [28]). More inequalities for the ratio $W_\nu(x)$ can be found in [15, 16, 29, 42, 47–53].

Using our main results, we can deduce some known inequalities for the ratio $W_\nu(x)$. For example, using the second inequality of (3.8) and Riccati Eq. (2.3) gives

$$\frac{W'_\nu(x)}{x} = \frac{x^2 + 2(\nu + 1)W_\nu(x) - W_\nu^2(x)}{x^2} < \frac{1}{\nu + 2},$$

which implies

$$\nu + 1 + \sqrt{\frac{\nu + 1}{\nu + 2}x^2 + (\nu + 1)^2} < W_\nu(x)$$

for $x > 0$ and $\nu > -1$. This inequality was first proven by Segura [49, Eq. (61)] (see also [28, Eq. (4.16)]). Since

$$\frac{W'_{\nu-1}(x)}{x} = \frac{I_\nu(x)^2 - I_{\nu+1}(x)I_{\nu-1}(x)}{I_\nu(x)^2},$$

the double inequality (3.8) implies the Turán type inequality

$$0 < I_\nu(x)^2 - I_{\nu+1}(x)I_{\nu-1}(x) < \frac{1}{\nu + 1}I_\nu(x)^2$$

for $x > 0$ and $\nu > -1$, which was first proven by Thiruvengatachar and Nanjundiah [54]. While the inequality (4.3), or equivalently, $0 < W'_{\nu-1}(x) < 1$ for $x > 0$ and $\nu \geq 1/2$, implies another Turán type inequality

$$0 < I_\nu(x)^2 - I_{\nu+1}(x)I_{\nu-1}(x) < \frac{1}{x}I_\nu(x)^2$$

for $x > 0$ and $\nu \geq 1/2$, which was due to Baricz [16, Eq. (2.5)].

In what follows, we list several new inequalities. The following corollary gives the lower and upper bounds for the function $W_\nu(x) - xW'_\nu(x)$ in the case of $\nu > -2$, which are combined by the inequalities (3.1), (3.2) and (3.3).

Corollary 4 *Let $\nu > -2$. The double inequality*

$$\min \left\{ \nu + \frac{1}{2}, 2\nu + 1 \right\} < W_\nu(x) - xW'_\nu(x) < \max \left\{ 2(\nu + 1), \frac{2(2\nu + 1)(\nu + 2)}{2\nu + 5} \right\} \quad (4.5)$$

holds for all $x > 0$.

Whence, we can deduce the following

Corollary 5 *Let $\nu > -2$. The double inequality*

$$x + \min \left\{ \nu + \frac{1}{2}, 2\nu + 1 \right\} < W_\nu(x) < x + \max \left\{ 2(\nu + 1), \frac{2(2\nu + 1)(\nu + 2)}{2\nu + 5} \right\} \quad (4.6)$$

holds for all $x > 0$.

Proof It follows from Corollary 1 that the first inequality of (4.6) for $\nu \geq -1/2$ and the second one of (4.6) for $\nu \geq -3/2$ hold for $x > 0$.

We next prove the inequality

$$x + 2\nu + 1 < W_\nu(x) \quad (4.7)$$

for $x > 0$ in the case of $-2 < \nu < -1/2$. In this case, the left hand side inequality of (4.5) becomes

$$2\nu + 1 < W_\nu(x) - xW'_\nu(x)$$

for $x > 0$, which is equivalent to

$$\left[-\frac{W_\nu(x)}{x} \right]' = -\frac{xW'_\nu(x) - W_\nu(x)}{x^2} > \frac{2\nu + 1}{x^2} = \left(-\frac{2\nu + 1}{x} \right)',$$

which in turn implies that the function $x \mapsto [W_\nu(x) - (2\nu + 1)]/x$ is strictly decreasing on $(0, \infty)$. It thus follows that

$$\frac{W_\nu(x) - (2\nu + 1)}{x} > \lim_{x \rightarrow \infty} \frac{W_\nu(x) - (2\nu + 1)}{x} = 1$$

for $x > 0$, and the inequality (4.7) follows.

Finally, we prove the inequality

$$W_\nu(x) < x + \frac{2(2\nu + 1)(\nu + 2)}{2\nu + 5} \quad (4.8)$$

for $x > 0$ in the case of $-2 < \nu < -3/2$. In this case the right hand side inequality of (4.5) changes to

$$W_\nu(x) - xW'_\nu(x) < \frac{2(2\nu+1)(\nu+2)}{2\nu+5}$$

for $x > 0$. Similarly, it implies that the function

$$x \mapsto \frac{1}{x} \left[W_\nu(x) - \frac{2(2\nu+1)(\nu+2)}{2\nu+5} \right]$$

is strictly increasing on $(0, \infty)$, which leads to (4.8), and the proof is completed. $\square$

Remark 9 In 1974 Näsell [55] proved a nice inequality

$$\left(1 + \frac{\nu}{x}\right) I_{\nu+1}(x) < I_\nu(x), \text{ or equivalently, } W_\nu(x) > x + \nu$$

for all $\nu > -1$ and $x > 0$. Now using the first inequality of (4.6) we have

$$W_\nu(x) > x + \nu + \min \left\{ \frac{1}{2}, \nu + 1 \right\} > x + \nu$$

for $x > 0$ if $\nu > -1$. This shows that our inequality is a refinement of Näsell's.

Taking into account the inequalities (4.4), (3.1) and (4.3) we obtain the following chain of inequalities.

Corollary 6 Let $\nu \geq -1/2$ and $x > 0$. The inequalities

$$0 < \frac{W_\nu(x) - 2(\nu+1)}{x} < W'_\nu(x) < 1 < \frac{W_\nu(x) - (\nu+1/2)}{x} \quad (4.9)$$

hold. The first and second inequalities of (4.9) also hold for $-3/2 \leq \nu < -1/2$.

Corollary 7 The inequality

$$W_\nu(x) > \nu + 1 + \sqrt{\left(x - \frac{1}{2}\right)^2 + \left(\nu + \frac{1}{2}\right)\left(\nu + \frac{3}{2}\right)} \quad (4.10)$$

holds for $x > 0$ and $\nu \geq -1/2$.

Proof Taking into account the inequality $W'_\nu(x) < 1$ and Riccati Eq. (2.3) gives

$$xW'_\nu(x) = x^2 + 2(\nu+1)W_\nu(x) - W_\nu^2(x) < x$$

for $x > 0$ and $\nu \geq -1/2$. This implies the required inequality. $\square$

Remark 10 Simpson and Spector [2, Theorem 2] proved that the inequality

$$W_\nu(x) > \nu + \frac{1}{2} + \sqrt{x^2 + \left(\nu + \frac{1}{2}\right)\left(\nu + \frac{3}{2}\right)} \quad (4.11)$$

holds for $x > 0$ and $\nu \geq 0$. Hornik and Grün [45] showed that this inequality is sharp and extended the range $\nu \geq 0$ to $\nu \geq -1/2$. It is easy to check that the inequality (4.10) is stronger than (4.11). In fact, let L and l denote the lower bounds given in (4.10) and (4.11), respectively. Then

$$\begin{aligned} \left(L - \nu - \frac{1}{2}\right)^2 - \left(l - \nu - \frac{1}{2}\right)^2 &= \sqrt{\left(x - \frac{1}{2}\right)^2 + \left(\nu + \frac{1}{2}\right)\left(\nu + \frac{3}{2}\right)} - \left(x - \frac{1}{2}\right) \\ &\geq \left|x - \frac{1}{2}\right| - \left(x - \frac{1}{2}\right) \geq 0 \end{aligned}$$

for $x > 0$ and $\nu \geq -1/2$.

Applying Jensen's inequality to the convex and concave functions given in Theorems 3 and 4, the following consequences are immediate.

Corollary 8 Let $\theta \geq 2$ and $\nu > -2$. The inequalities

$$\begin{aligned} 0 &< \frac{W_\nu(x) + W_\nu(y)}{2} - W_\nu\left(\frac{x+y}{2}\right) < \frac{1}{8} \frac{(x-y)^2}{\nu+2}, \\ W_\nu\left(\frac{x+y}{2}\right) &< \frac{W_\nu(x) + W_\nu(y)}{2} < W_\nu\left(\left(\frac{x^\theta + y^\theta}{2}\right)^{1/\theta}\right) \end{aligned}$$

hold for $x, y > 0$ with $x \neq y$ and $\nu \geq -1/2$, where the right hand side inequalities hold for $\nu > -2$.

5 Applications

5.1 An answer to Gaunt's open problem

The variance-gamma (VG) distribution introduced into the financial literature in [56], also known as the generalized Laplace or Bessel function distribution, is a continuous statistical distribution defined and supported on the set of real numbers. For variance-gamma approximation via Stein's method [57] (see also [58]), it is crucial to obtain the uniform bounds for four expressions of the type listed in [59, Eq. (1.1), (1.2)]. Gaunt has obtained uniform bounds for them, where the bound for the third follows from [59, Eq. (2.30)], which in turn depends on the upper bound for the integral

$$\int_0^x e^{\beta t} t^\nu I_\nu(t) dt \text{ for } \nu > -\frac{1}{2}, \beta \in (-1, 0).$$

For this reason, he proved Theorem 2.7 in [59] by using the inequality

$$I_{\nu+1}(x) < (1 - a_\nu) I_\nu(x) + a_\nu I_{\nu+2}(x) \quad (5.1)$$

for $x > 0$, where a_ν is the largest number in the interval $[0, 1]$. Gaunt claimed that “the inequality has not been studied in the literature” and left for readers as an open problem.

Open Problem 1 ([59, Open Problem 2. 11]) *Let $\nu > -1/2$. Establish lower and upper bounds for a_ν that improve on the trivial estimate $0 \leq a_\nu \leq 1$, where a_ν satisfies the inequality (5.1).*

We now give an answer to the above open problem using Corollary 1.

Proposition 1 *Let $\nu \geq -1/2$ and $0 < a_\nu < 1$. The inequality (5.1) holds for $x > 0$ if and only if $0 < a_\nu \leq (\nu + 1/2) / (2\nu + 2)$.*

Proof The inequality (5.1) is equivalent to

$$a_\nu < \frac{I_\nu(x) - I_{\nu+1}(x)}{I_\nu(x) - I_{\nu+2}(x)} := q_\nu(x)$$

for $x > 0$. Using the recurrence formula (1.5) gives

$$q_\nu(x) = \frac{I_\nu(x) - I_{\nu+1}(x)}{2(\nu+1)x^{-1}I_{\nu+1}(x)} = \frac{W_\nu(x) - x}{2(\nu+1)}.$$

The necessary condition follows from

$$0 < a_\nu \leq \lim_{x \rightarrow \infty} q_\nu(x) = \frac{\nu + 1/2}{2(\nu + 1)}.$$

By Corollary 1 (i) we deduce that the function $q_\nu(x)$ is strictly decreasing on $(0, \infty)$ with

$$q_\nu(0) = 1 \text{ and } q_\nu(\infty) = \lim_{x \rightarrow \infty} q_\nu(x) = \frac{\nu + 1/2}{2(\nu + 1)}.$$

If $0 < a_\nu \leq (\nu + 1/2) / (2\nu + 2)$, then by Corollary 1 (i), the function $q_\nu(x)$ is strictly decreasing on $(0, \infty)$, and therefore,

$$a_\nu \leq \frac{\nu + 1/2}{2(\nu + 1)} = \lim_{x \rightarrow \infty} q_\nu(x) < q_\nu(x)$$

for $x > 0$, thereby completing the proof. $\square$

Remark 11 Taking $a_\nu = (\nu + 1/2) / (2\nu + 2)$, Gaunt’s Theorem 2.7 in [59] can be improved; correspondingly, Theorem 3.2 (iv) [59] can also be slightly improved.

Remark 12 Gaunt [59, Open Problem 2.12] also proposed an analogous problem for the modified Bessel function $K_\nu(x)$: Let $\nu > -1/2$. Establish lower and upper bounds for b_ν that improve on the trivial estimate $0 \leq b_\nu \leq 1$ such that, for all $x > 0$, the inequality

$$K_{\nu+1}(x) < b_\nu K_\nu(x) + (1 - b_\nu) K_{\nu+2}(x)$$

holds. Using Proposition 2 in [43] we easily get the best constant $b_\nu = (2\nu + 3) / (4\nu + 4)$.

5.2 An answer to Baricz and Neuman's conjecture

Násell [42] proved that the function $x \mapsto x^{-\nu} e^{-x} I_\nu(x)$ is completely monotonic on $(0, \infty)$ for $\nu \geq -1/2$. This implies that the function $x \mapsto \mathcal{I}_\nu(x) = 2^\nu \Gamma(\nu + 1) x^{-\nu} I_\nu(x)$ is log-convex on $(0, \infty)$ for $\nu \geq -1/2$, since a completely monotonic function is log-convex. Other proofs of this result can be found in [14, 46, 60]. In 2007, Baricz and Neuman [14] made the following conjecture.

Conjecture 2 *The function $x \mapsto \mathcal{I}_\nu(x) = 2^\nu \Gamma(\nu + 1) x^{-\nu} I_\nu(x)$ is strictly log-convex on $\mathbb{R}$ for all $\nu > -1$ while the function $x \mapsto I_\nu(x)$ is strictly log-concave on $(0, \infty)$ for $\nu > 0$.*

We now give an answer to this conjecture.

Proposition 2 *Let $\nu > -1$. The following statements are true.*

- (i) *The function $x \mapsto \mathcal{I}_\nu(x) = 2^\nu \Gamma(\nu + 1) x^{-\nu} I_\nu(x)$ is log-convex on $(0, \infty)$ if and only if $\nu \geq -1/2$, and it is log-convex then log-concave on $(0, \infty)$ if $\nu \in (-1, -1/2)$.*
- (ii) *When $\nu > 0$, the function $x \mapsto I_\nu(x)$ is log-concave then log-convex on $(0, \infty)$.*

Proof Let

$$h_{p,\nu}(x) = x W'_\nu(x) - W_\nu(x) + p.$$

Differentiation yields

$$\begin{aligned} x^2 [\ln \mathcal{I}_\nu(x)]'' &= x W'_{\nu-1}(x) - W_{\nu-1}(x) + 2\nu = h_{2\nu,\nu-1}(x), \\ x^2 [\ln I_\nu(x)]'' &= x W'_{\nu-1}(x) - W_{\nu-1}(x) + \nu = h_{\nu,\nu-1}(x). \end{aligned}$$

It is easy to check that

$$\begin{aligned} h_{2\nu,\nu-1}(0) &= 0 \text{ and } h_{2\nu,\nu-1}(\infty) = \lim_{x \rightarrow \infty} h_{2\nu,\nu-1}(x) = \nu + \frac{1}{2}; \\ h_{\nu,\nu-1}(0) &= -\nu \text{ and } h_{\nu,\nu-1}(\infty) = \lim_{x \rightarrow \infty} h_{\nu,\nu-1}(x) = \frac{1}{2}. \end{aligned}$$

(i) Clearly, the necessary condition for $[\ln \mathcal{I}_\nu(x)]'' > 0$ is: $\nu \geq -1/2$. Conversely, if $\nu \geq -1/2$, i.e. $\nu - 1 \geq -3/2$, then by the left hand side inequalities of (3.1) and (3.2) $h_{2\nu,\nu-1}(x) \geq -2\nu + 2\nu = 0$ for $x > 0$, which proves the sufficiency.

When $\nu \in (-1, -1/2)$, by Theorem 1 (ii) it is seen that the function $h_{2\nu, \nu-1}(x)$ is first increasing then decreasing and finally increasing on $(0, \infty)$. This together with $h_{2\nu, \nu-1}(0) = 0$ and $h_{2\nu, \nu-1}(\infty) = \nu + 1/2 < 0$ indicates that the sign of $h_{2\nu, \nu-1}(x)$ changes from positivity to negativity, and hence, the function $x \mapsto \ln \mathcal{I}_\nu(x)$ is convex then concave on $(0, \infty)$.

(ii) Similarly, we can prove the function $x \mapsto \ln I_\nu(x)$ is convex then concave on $(0, \infty)$. $\square$

Remark 13 Using the same method as in the above proof, we can show the necessary and sufficient conditions for which the function $\phi_{p, \nu}(x) = x^p I_\nu(x)$ is log-convex or log-concave on $(0, \infty)$ as follows.

- (i) If $\nu \geq -1/2$, then the function $\phi_{p, \nu}(x)$ is log-convex on $(0, \infty)$ if and only if $p \leq -\nu$.
- (ii) If $\nu \geq 1/2$, then the function $\phi_{p, \nu}(x)$ is log-concave on $(0, \infty)$ if and only if $p \geq 1/2$.
- (iii) If $\nu \geq -1/2$ and $-\nu < p < 1/2$, then the function $\phi_{p, \nu}(x)$ is log-concave then log-convex on $(0, \infty)$.

6 Conclusions

In this paper, using the monotonicity rules for a ratio of power series and an elementary auxiliary function we proved several convexity results:

- For $\nu > -2$, the functions $x \mapsto W_\nu(x)$ and $x \mapsto [W_\nu(x) - (\nu + 1/2)]/x$ are both strictly convex on $(0, \infty)$ if and only if $\nu \geq -1/2$.
- For $\theta \geq 2$ and $\nu > -2$, the functions $x \mapsto W_\nu(x^{1/\theta})$ and $x \mapsto W_\nu(x) - x^2/(2\nu+4)$ are both strictly concave on $(0, \infty)$, while $x \mapsto [W_\nu(x) - 2(\nu+1)]/x$ is strictly concave on $(0, \infty)$ if and only if $\nu \geq -3/2$.

As direct consequences of the above main results, we describe the monotonic pattern of the difference $W_\nu(x) - x$ on $(0, \infty)$ for ν in the intervals $[-1/2, \infty)$, $[-3/2, -1/2)$ and $(-2, -3/2)$, and establish several new inequalities for the ratio $W_\nu(x)$. Finally, as applications, we give answers to an open problem and a conjecture.

Finally, let us recall that an infinitely differentiable function $f : I \rightarrow [0, \infty)$ is called a Bernstein function on an interval I if f' is completely monotonic on I (see [61, Definition 3.1]). Now, inspired the facts that $W_\nu(x), W'_\nu(x) > 0$ and $[W_\nu(x^{1/\theta})]'' < 0$ for $x > 0$ and $\nu > -1$ with $\theta \geq 2$, we propose a conjecture as follows.

Conjecture 3 Let $\tau \in (0, 1/2]$ and $\nu > -1$. The function $x \mapsto W_\nu(x^\tau)$ is a Bernstein function on $(0, \infty)$.

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